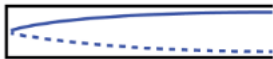


Q14

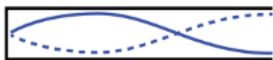
Let  $L$  be the length of the bugle,  $v$  be the speed of sound in air,  $\lambda$  be the wavelength.

92 Hz corresponds to that shown below:



$$4L = \lambda_1 = v/92, \quad v = 368L$$

The next higher frequency corresponds to that shown below:



$$3\lambda_2/4 = L \text{ or } 3(v/f_2) = 4L \text{ or } 3(368L/f_2) = 4L, \quad f_2 = 276 \text{ Hz}$$

The next higher frequency corresponds to that shown below:



$$5\lambda_3/4 = L, \quad 5(v/f_3) = 4L, \quad 5(368L/f_3) = 4L, \quad f_3 = 460 \text{ Hz}$$

Q15